

Lab 01: The Rental Refund Crisis

Exercise 1: Testing the "All or Nothing"

Query Query History

```
1 BEGIN;
2     UPDATE rental SET return_date = NOW() WHERE rental_id=2;;
3     INSERT INTO payment (customer_id, staff_id, rental_id, amount,
4 payment_date) VALUES(459, 1, 2, 99.99, 'String');
5 COMMIT;
6
```

Data Output Messages Notifications

ERROR: invalid input syntax for type timestamp: "String"
LINE 4: payment_date) VALUES(459, 1, 2, 99.99, 'String');
^

SQL state: 22007
Character: 170

Query Query History

```
1 SELECT * from rental WHERE rental_id = 2
2
```

Data Output Messages Notifications

Showing rows: 1 to 1

Page No

	rental_id [PK] integer	rental_date timestamp without time zone	inventory_id integer	customer_id smallint	return_date timestamp without time zone	staff_id smallint	last_update timestamp without time zone
1	2	2005-05-24 22:54:33	1525	459	2005-05-28 19:40:33	1	2006-02-16 02:30:53

The return_date column is not updated because the query INSERT INTO runs into an error. This violates the Atomicity property

Exercise 2: The "Flexible" Staff Task

Query Query History

```
1 SELECT * from staff
2
```

Data Output Messages Notifications

Showing rows: 1 to 2 Page No: 1 of 1

	staff_id [PK] integer	first_name character varying (45)	last_name character varying (45)	address_id smallint	email character varying (50)	store_id smallint	active boolean	username character varying (16)	password character varying (40)
1	1	Mike	Hillyer	3	Mike.Hillyer@sakilastaff.com	1	true	Mike	8cb2237d0679ca88db6464eac60da96345
2	2	Jon	Stephens	4	Jon.Stephens@sakilastaff.c...	2	true	Jon	8cb2237d0679ca88db6464eac60da96345

```
18 BEGIN;
19     INSERT INTO staff (first_name, last_name, address_id, email, store_id, username)
20     VALUES ('Tom', 'Hanks', 6, 'TH@gmail.com', 1, 'Tom');
21     SAVEPOINT add_staff;
22
23     UPDATE staff SET store_id = 3 WHERE email = 'TH@gmail.com';
24     ROLLBACK TO add_staff;
25
26 COMMIT;
```

Data Output Messages Notifications

COMMIT

Query returned successfully in 53 msec.

- Although there is no store_id = 3, the query continues to run thanks to the ROLLBACK
- When it fails, Rollback to Savepoint and assign them back to Store 1

Query Query History

```
1 SELECT * from staff
2
```

Data Output Messages Notifications

Showing rows: 1 to 3 Page No: 1 of 1

	staff_id [PK] integer	first_name character varying (45)	last_name character varying (45)	address_id smallint	email character varying (50)	store_id smallint	active boolean	username character varying (16)	password character varying (40)
1	1	Mike	Hillyer	3	Mike.Hillyer@sakilastaff.com	1	true	Mike	8cb2237d0679ca88db6464eac60
2	2	Jon	Stephens	4	Jon.Stephens@sakilastaff.c...	2	true	Jon	8cb2237d0679ca88db6464eac60
3	6	Tom	Hanks	6	TH@gmail.com	1	true	Tom	[null]